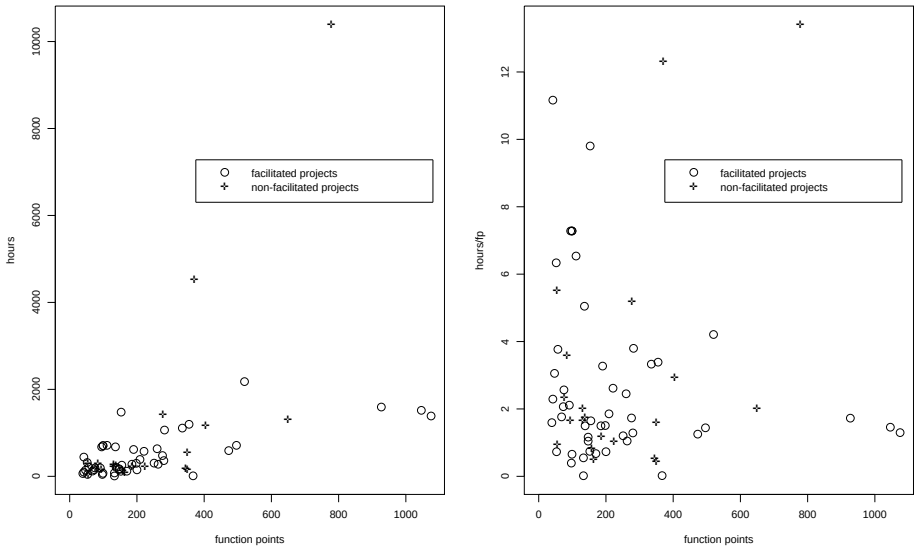


(a) Required effort

(b) Required effort per function point

Figure 4.2: Effect of requirements engineering technique on required effort for requirements engineering.

If we however plot the productivity in requirements engineering, in terms of hours spent by ICT personnel per function point against the size of the projects, a pattern does appear. When we examine only the small software development projects some projects using facilitated workshops perform worse than their traditional Method/1 counterparts. When we examine large projects the opposite can be observed: here Method/1 projects under-perform while facilitated workshops are more productive.

To determine whether small projects are really better off without facilitated workshops and large project are really better off with facilitated workshops, the statistical significance of the pattern is tested with an ANOVA analysis. If the effect caused by the project size (in function points) on the effectiveness of the requirements engineering technique (facilitated workshops or one-on-one interviews) cannot be explained by chance alone, the interaction between the factors technique and size should be significant.

The dependent variable productivity is severely left-skewed. To obtain an approximately normally distributed variable, the dependent variable has been transformed with an square root transformation. The transformed variable has been used in the ANOVA analysis.

When testing the interaction effect between requirements engineering technique and project size with an ANOVA test, the interaction effect (technique:size in Table 4.1) turns out to be significant well beyond the 0.05 α-level. The first line (labelled 'technique'), reflects the influence of the requirements engineering method on total effort. The second